LIGHTWEIGHT ML-BASED INTRUSION DETECTION FOR CAMPUS NETWORKS

CS1140: Introduction to Cyber Security

Course Project



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1. Project Overview

Campus networks connect a large number of users and devices, including student laptops, faculty systems, laboratory computers, servers, IoT devices, and other network-connected resources. Due to the large and diverse number of devices, campus networks can become targets for various cyber threats such as port scanning, denial-of-service attacks, brute-force attempts, suspicious network connections, and unauthorized access.

Traditional Intrusion Detection Systems (IDS) often rely heavily on predefined signatures or require considerable computational resources for continuous network monitoring. Such approaches may be difficult to deploy efficiently in resource-constrained campus environments and may also struggle to identify previously unseen or slightly modified attack patterns.

This project proposes a **Lightweight Machine Learning-Based Intrusion Detection System** designed specifically for campus networks. The system will analyze network traffic and extract relevant characteristics such as packet statistics, connection duration, protocol information, source and destination details, packet sizes, and traffic frequency. These features will be provided to a lightweight machine learning model that classifies network activity as **normal or potentially malicious**.

The primary objective is to develop a detection system that provides a balance between **detection accuracy, computational efficiency, and real-time monitoring capability**. The proposed system will be evaluated using suitable network intrusion datasets and, where feasible, controlled network traffic generated in a laboratory environment.

The final system will provide security alerts and basic traffic statistics, allowing administrators or users to identify suspicious network activity and investigate potentially malicious connections.

2. Core Features

2.1 Lightweight ML-Based Detection

The system will use computationally efficient machine learning algorithms to classify network traffic. Models such as **Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), or LightGBM** can be evaluated to identify an appropriate balance between accuracy and resource consumption.

2.2 Network Traffic Feature Extraction

The system will extract meaningful features from network traffic, including source and destination IP information, source and destination ports, network protocol, packet size, number of packets, connection duration, flow frequency, forward and backward packet statistics, TCP flag information, bytes transferred, and connection-based traffic characteristics. These features will be processed before being provided to the machine learning model.

2.3 Normal and Malicious Traffic Classification

The trained model will classify network flows into normal or suspicious categories. Depending on the selected dataset and model, multiple intrusion categories can also be considered, such as denial-of-service attacks, port scanning, brute-force attacks, botnet-related traffic, network probing, and other abnormal network behaviour.

2.4 Real-Time Network Monitoring

A lightweight monitoring component will observe network traffic and pass relevant traffic features to the trained model. When suspicious behaviour is detected, the system will generate an alert containing

important information about the detected traffic.

2.5 Risk-Based Alerting

Instead of simply displaying a binary result, the system can assign a confidence or risk score to detected traffic. Higher-risk connections can be highlighted for further investigation.

2.6 Performance and Resource Evaluation

Since the project focuses on a **lightweight** IDS, the system will be evaluated not only on detection performance but also on computational requirements. The evaluation can consider detection accuracy, precision, recall, F1-score, false-positive rate, prediction latency, CPU usage, and memory consumption.

2.7 Security Monitoring Interface

A simple terminal-based or lightweight dashboard interface will display current network activity, detected intrusions, source and destination information, attack category, risk/confidence score, detection statistics, and timestamped alerts. The interface will also maintain security logs for later analysis.

3. Technology Stack

Programming Language: Python

Packet Inspection & Networking: Scapy, PyShark, Wireshark, Tcpdump

Data Processing & Feature Extraction: Pandas, NumPy

Machine Learning: Scikit-Learn, Logistic Regression, Decision Tree, Random Forest, KNN, LightGBM

Data Visualization / Monitoring: Matplotlib and/or a lightweight terminal interface

Logging: CSV / JSON

Dataset: A suitable publicly available network intrusion detection dataset such as CIC-IDS2017 or another appropriate benchmark dataset.

Testing Environment: Linux/Windows systems and controlled virtual network environments using VirtualBox where required.

4. Proposed Workflow

The proposed workflow follows an end-to-end pipeline from network traffic collection to intrusion alert generation.

1	Network Traffic / Dataset
2	Traffic Collection
3	Data Cleaning & Preprocessing
4	Feature Extraction
5	Feature Selection / Reduction
6	Lightweight ML Model
7	Normal / Intrusion Classification

8	Risk Scoring & Alert Generation
9	Logging & Monitoring Interface

The workflow will begin with the collection or selection of network traffic data. The collected data will be cleaned and transformed into a suitable format for machine learning. Relevant traffic features will then be extracted and selected to reduce unnecessary computational overhead.

Different lightweight machine learning models will be trained and compared. The selected model will subsequently be integrated into the monitoring component. During operation, incoming traffic will be converted into the required feature representation and passed to the model for classification.

If suspicious behaviour is detected, the system will generate an alert and record the event for further analysis.

5. Project Objectives

- To understand common network attacks relevant to campus environments.
- To develop a machine learning-based intrusion detection approach.
- To identify network traffic features that are useful for intrusion detection.
- To compare lightweight machine learning algorithms.
- To reduce unnecessary computational and memory requirements.
- To achieve effective detection while minimizing false positives.
- To develop a basic real-time monitoring and alerting mechanism.
- To evaluate the proposed IDS using standard machine learning and cybersecurity metrics.

6. Timeline

Considering that this project is undertaken as a course project, the proposed development timeline is approximately **10 weeks**.

Week 1	Literature review, cybersecurity threat analysis, and IDS research.
Week 2	Dataset selection, environment setup, and initial network traffic analysis.
Week 3	Dataset preprocessing, cleaning, feature extraction, and feature analysis.
Week 4	Selection and implementation of lightweight machine learning algorithms.
Week 5	Model training and initial performance evaluation.
Week 6	Model comparison, feature selection, hyperparameter tuning, and false-positive analysis.
Week 7	Integration of the selected ML model with the network monitoring component.
Week 8	Development of the monitoring interface, alert mechanism, and security logging.
Week 9	Real-time/controlled network testing, performance measurement, and edge-case investigation.
Week 10	Final testing, debugging, documentation, performance analysis, and project polishing.

7. Expected Outcome

The expected outcome of this project is a **lightweight and practical machine learning-based Intrusion Detection System** capable of identifying suspicious network activity in a campus-network environment.

The system is expected to demonstrate that machine learning can be used to detect network intrusions while maintaining relatively low computational requirements. The project will also provide a comparison of different ML models based on detection performance and resource efficiency.

The final prototype will provide basic real-time monitoring, intrusion alerts, risk information, and security logs that can assist in identifying potentially malicious network behaviour.

8. Conclusion

This project combines **machine learning and cybersecurity** to develop a lightweight Intrusion Detection System suitable for campus-network environments. Rather than relying exclusively on predefined attack signatures, the proposed system will learn patterns from network traffic and use these patterns to identify abnormal or malicious activity.

The project emphasizes the practical balance between **security, accuracy, speed, and resource consumption**. Through feature engineering, lightweight machine learning models, controlled testing, and performance evaluation, the system aims to provide an efficient approach for detecting common network intrusions.

The final deliverable will be a functional prototype capable of analysing network traffic, classifying suspicious activity, generating alerts, and maintaining logs for security analysis. The exact dataset, machine learning model, features, and implementation details may be modified during development based on experimental results and project requirements.